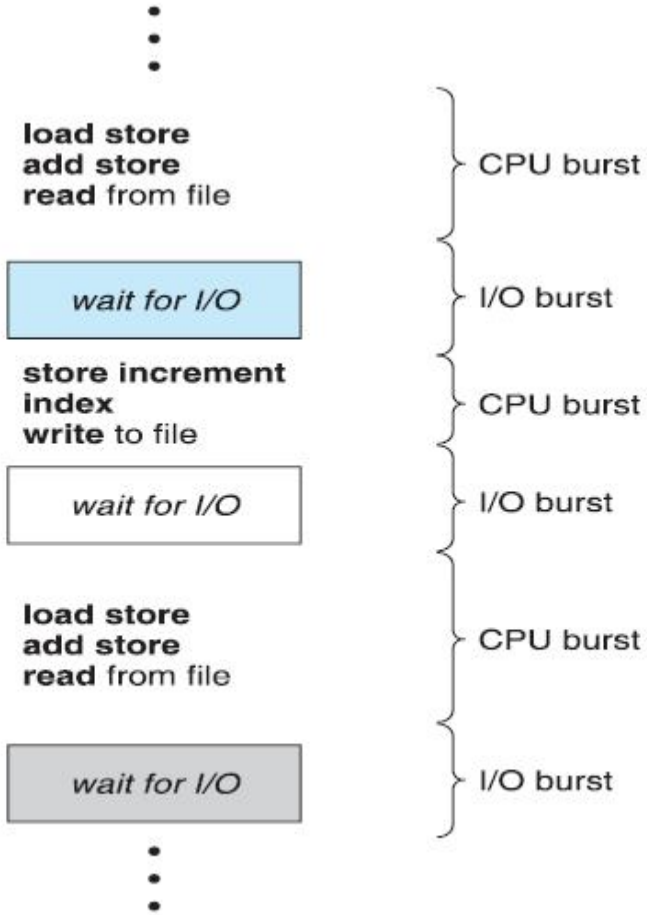
Process scheduling

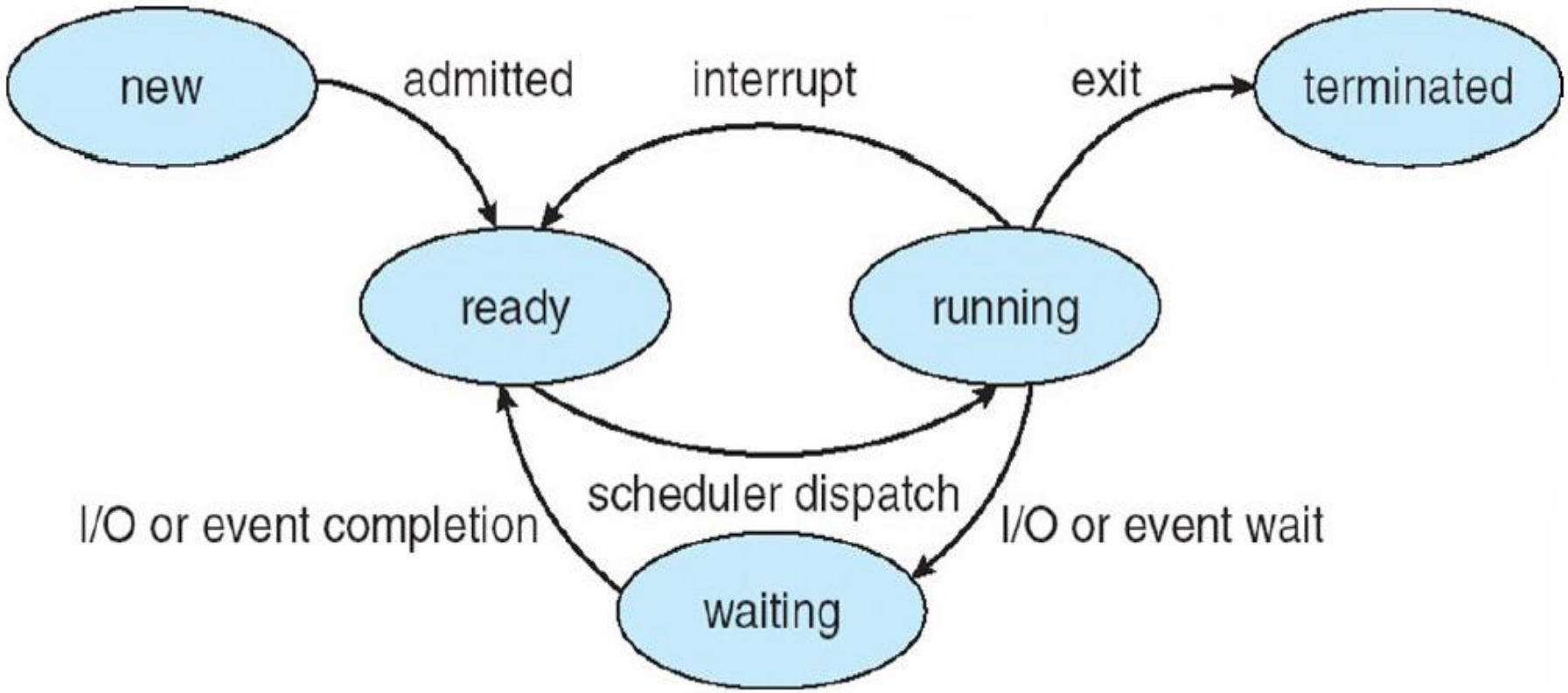
- Maximize CPU utilization.
- Scheduling – function of operating system.

CPU-I/O Burst cycle

- Process execution - cycle of CPU execution and I/O wait.
- CPU burst followed by I/O burst
- Final CPU burst ends with system request to terminate execution.



Process states



Scheduling circumstances

- When a process switches from the running state to the waiting state.
- When a process switches from the running state to the ready state.
- When a process switches from the waiting state to the ready state.
- When a process terminates.

Preemptive Scheduling

Preemptive scheduling is used when a process switches from running state to ready state (2) or from waiting state to ready state (3).

The resources are allocated to the process for the limited amount of time and then is taken away, and the process is again placed back in the ready queue if that process still has CPU burst time remaining.

That process stays in ready queue till it gets next chance to execute.

Non-Preemptive Scheduling

Non-preemptive Scheduling is used when a process terminates (4), or a process switches from running to waiting state (1).

In this scheduling, once the resources is allocated to a process, the process holds the CPU till it gets terminated or it reaches a waiting state.

No other process interrupt a process running in middle of the execution.

Instead, it waits till the process complete its CPU burst time and then it can allocate the CPU to another process.

Scheduling criteria

CPU utilization – keep the CPU as busy as possible

Throughput – # of processes that complete their execution per time unit

Turnaround time – amount of time to execute a particular process

Waiting time – amount of time a process has been waiting in the ready queue

Response time – amount of time it takes from when a request was submitted until the first response is produced, not output

Optimization Criteria

CPU utilization – Maximize

Throughput – Maximize

Turnaround time – Minimize

Waiting time – Minimize

Response time – Minimize

Scheduling algorithms

- First-come, First-Served
- Shortest-Job First
- Priority Scheduling
- Round-Robin
- Multilevel queue
- Multilevel Feedback Queue

First-Come, First-Served Scheduling

- The process that requests the CPU first is allocated first.
- FCFS algorithm in non preemptive
- Convoy effect – all the other processes wait for one big process to get off the CPU.

First-Come, First-Served Scheduling

Process id	Arrival time	Burst time/CPU execution time
P1	0	2
P2	1	3
P3	2	5
P4	3	4
P5	4	6

First-Come, First-Served Scheduling

Process id	Arrival time	Burst time/CPU execution time
P1	0	2
P2	1	3
P3	2	5
P4	3	4
P5	4	6

Average Turnaround time =2+4+8+11+16/5= 8.2

Average Waiting Time = 0+1+3+7+10/5=4.2

First-Come, First-Served Scheduling

PROCESS	ARRIVAL TIME	BURST TIME
P1	0	8
P2	1	4
P3	2	9
P4	3	5

Average Turnaround time =61/4=15.25
Average Waiting Time = 35/4=8.75
Throughput=

Shortest Job First Scheduling

- When the CPU is available, it is assigned to the process that has the smallest CPU burst.
- If the CPU bursts of two processes are same, then FCFS is used.
- SJF can be preemptive or non preemptive

Shortest Job First Scheduling

- SJF using preemptive or non preemptive

Process Id	Arrival time	Burst time
P1	3	1
P2	1	4
P3	4	2
P4	0	6
P5	2	3

Shortest Job First Scheduling

Process Id	Arrival time	Burst time
P1	3	1
P2	1	4
P3	4	2
P4	0	6
P5	2	3

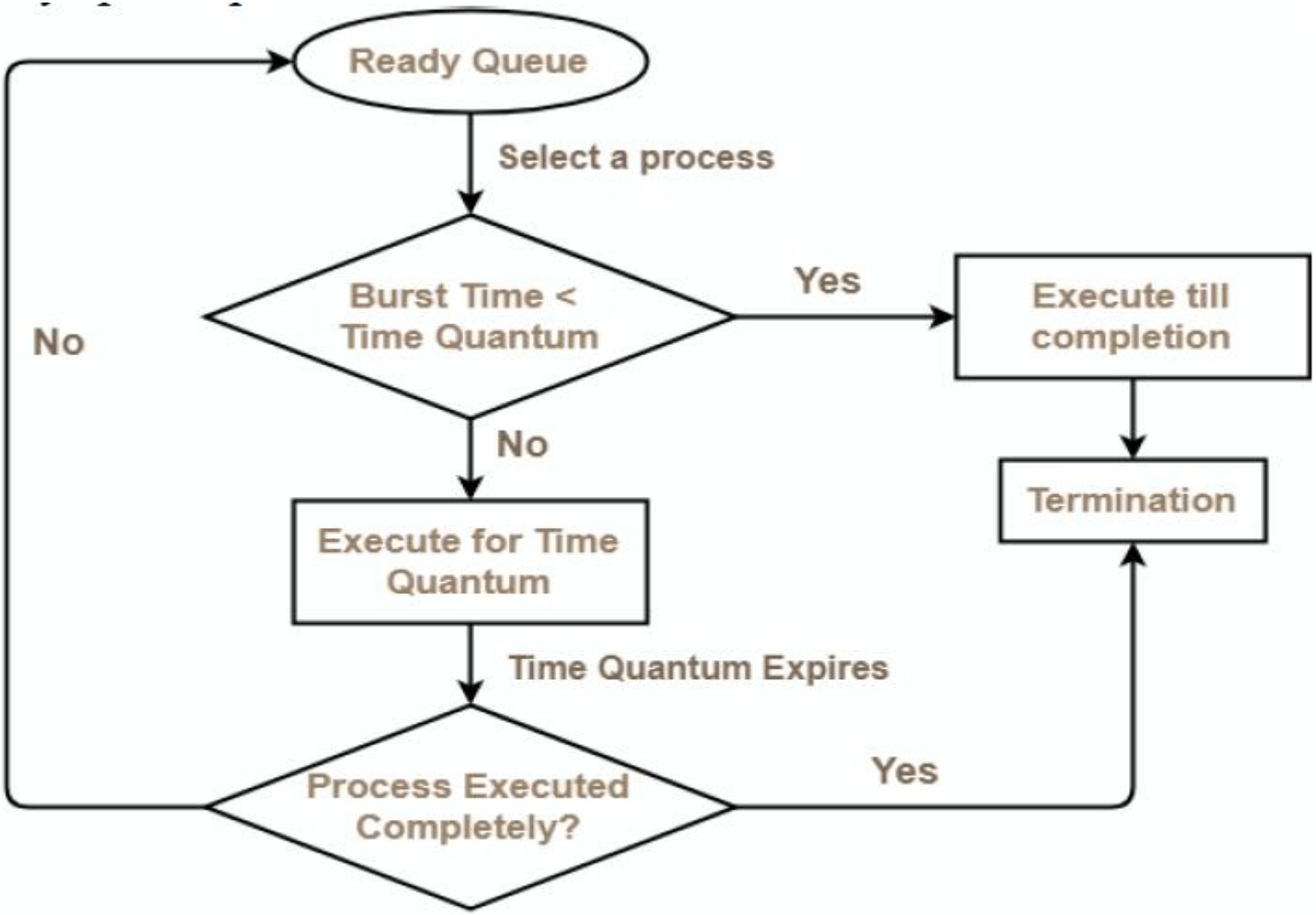
Round Robin Scheduling

CPU is assigned to the process on the basis of FCFS for a fixed amount of time.

- This fixed amount of time is called as time quantum or time slice.

- After the time quantum expires, the running process is preempted and sent to the ready queue.

- Then, the processor is assigned to the next arrived process.



Round Robin Scheduling

Round Robin Scheduling

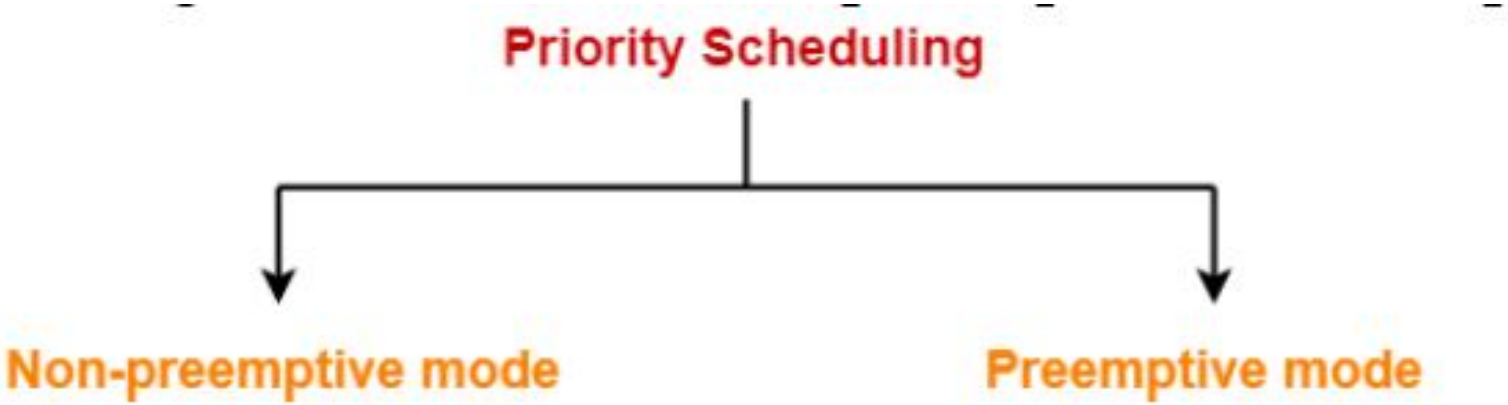
Process Id	Arrival time	Burst time
P1	0	5
P2	1	3
P3	2	1
P4	3	2
P5	4	3

Priority Scheduling

- Out of all the available processes, CPU is assigned to the process having the highest priority.

- In case of a tie, it is broken by FCFS Scheduling.

Priority Scheduling



Priority Scheduling (Non-preemptive)

Process Id	Arrival time	Burst time	Priority
P1	0	4	2
P2	1	3	3
P3	2	1	4
P4	3	5	5
P5	4	2	5

Priority Scheduling (Preemptive)

Process Id	Arrival time	Burst time	Priority
P1	0	4	2
P2	1	3	3
P3	2	1	4
P4	3	5	5
P5	4	2	5

- Turn Around time = Exit time – Arrival time
- Waiting time = Turn Around time – Burst time